PakCareer AI

Data-driven degree guidance for Pakistani college graduates

Deliverable 1: Problem Definition & Data Foundations (Track A)

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Abstract

Every year, students finishing intermediate or A-Levels in Pakistan choose a degree that shapes their whole career, and the choice is very often made by their parents, who default to a small set of "safe" professions. PakCareer AI sorts a student into one of seven degree families ("houses") from their interests and personality, and backs the recommendation with Pakistan-specific labour-market statistics the student can show their family. This report covers Phase 1 (problem definition) and Phase 2 (data acquisition and preprocessing). We collected five sources: 145,828 RIASEC interest-test responses (training data), the O*NET 31.0 occupation database, Pakistan Labour Force Survey indicators from ILOSTAT, World Bank indicators, and 10,999 live Pakistani job postings scraped from Rozee.pk. We also evaluated ten supplementary Kaggle career datasets and excluded all of them from training: six have random labels, one leaks the target and three are too small or unlabelled. After cleaning, the modelling table holds 81,256 labelled respondents with 30 engineered, scaled and encoded features. A preliminary gradient-boosting baseline places the student's actual field among its top three houses 86% of the time.

1 Phase 1: Problem Definition

1.1 Problem statement

In Pakistan the decision of what to study after intermediate (FSc, ICS, ICom, FA) or A-Levels is frequently taken by the family rather than the student. Medicine and engineering in particular are treated as default "respectable" careers, regardless of whether the student is interested in or suited to them. The result is students who are unmotivated in their degree, switch fields mid-way, or graduate into a career they did not want.

Choosing "any degree" is also no longer a safe bet. The Labour Force Survey data we collected (Section **??**) shows that in 2025 the unemployment rate among Pakistanis with advanced (degree-level) education was **11.1%**, against **7.0%** for the workforce as a whole, and it peaked above 16% in 2015 and 2021. For women with a degree it was **23.9%**, compared with 6.8% for men. A degree that does not fit the student, in a field the market does not need, is costly.

Students who disagree with their parents usually have nothing but their own opinion to argue with. Career counselling is rare in Pakistani colleges, and generic online personality quizzes give no evidence a parent would take seriously.

PakCareer AI addresses both halves of the problem:

1. **Fit:** the student completes a short assessment of interests (the 48-item RIASEC inventory) and personality (TIPI). A trained classifier estimates how well they fit each of seven degree families: Medicine & Health, Engineering, Computer Science & IT, Business & Commerce, Law, Arts, Humanities & Social Sciences, and Pure Sciences.
2. **Evidence:** the recommendation comes with a statistical report built from Pakistani data, covering earnings by sector, unemployment by education level, and current job demand per field and city. The student can use it to make an informed case to their parents.

1.2 Target audience

- **Primary users:** Pakistani students aged roughly 16–20 who are finishing intermediate or A-Levels, or are in the first year of a degree and considering a switch.
- **Secondary users:** their parents, who read the statistical report, and teachers or college counsellors who want a structured, evidence-based starting point for a career conversation.

1.3 AI justification

A rule-based system would encode Holland's theory directly, e.g. "highest score Realistic ⇒ Engineering". Our data shows why that is not enough:

- **High-dimensional, overlapping inputs.** Each student is described by 58 item responses (or 30 engineered features). Houses overlap heavily. Engineering and CS & IT are *both* highest on Realistic, and Medicine and Pure Sciences are *both* highest on Investigative (Figure **??**). They differ only in the combination of several weaker signals (Social, Conventional, Agreeableness, . . .). Hand-writing thresholds for every combination of 30 features is infeasible and would be arbitrary.
- **Noisy, probabilistic relationship.** People with the same profile end up in different fields. A useful system must output a *probability* of fit for every house, not a single hard rule, so the student and parents can see second and third choices. A learned model calibrated on 81k labelled examples gives this directly.
- **Learned weights instead of guessed ones.** How much should vocabulary, openness or age count relative to interest scores? A model estimates these weights from data and can be evaluated (accuracy, macro-F1) on held-out students. A rule system cannot be validated the same way.
- **Adaptable to Pakistan.** The training data is international (53% US). With an ML pipeline, the Pakistani survey responses we are collecting can be used to re-weight or fine-tune the model and to measure how well it transfers. A fixed rule set would have to be rewritten by hand.
- **Bias can be measured.** Because predictions are learned, we can audit them, for example by checking whether the model under-recommends Engineering to women (Figure **??**), and correct with re-weighting.

1.4 Proposed system

1. The student takes the assessment in a web app (RIASEC items, TIPI, a few background questions).
2. The same preprocessing pipeline fitted in this deliverable (`preprocessor.joblib`) transforms the answers. A classifier (Phase 3: logistic regression, random forest, gradient boosting and an MLP will be compared) outputs a probability for each house.
3. The report generator combines the top houses with the cleaned Pakistan statistics into a downloadable report for parents.

2 Phase 2: Data Acquisition

2.1 Sources

All raw data was collected by one reproducible script, `data_scripts/collect_data.py` (Python standard library only). It writes `data/raw/collection_log.json`, recording the URL, file, row count, size and timestamp of every download as evidence of collection. Table **??** summarises the sources.

Table 1: Data sources collected on 18 October 2026.

Source	Method	Size	Role
Open Psychometrics RIASEC test	ZIP download	145,828 × 94	**Training data**: 48 interest items, TIPI, vocabulary check, demographics, free-text university major (target)
O*NET 31.0 (US Dept. of Labor)	ZIP download	21 tables, ∼1,016 occupations	RIASEC profile, education and skills of each occupation; used to design per-house tasks
ILOSTAT, Pakistan LFS	SDMX REST API	9 indicators, 36,954 rows	Earnings by sector/occupation/education, unemployment by education, employment, all by sex
World Bank	REST API	16 indicators, 1990–2025	Enrolment, unemployment, inflation, exchange rate
Rozee.pk job postings	Web scraping	10,999 postings	Current Pakistani job demand per field and city; degree and experience asked for
Pakistani student survey	Google Form (ours)	ongoing	Local validation data with the same RIASEC/TIPI items plus parental-pressure questions
ESCO (European Commission)	download	3,039 occupations	Occupation taxonomy with ISCO codes (links job titles to ILO occupation groups)
Career knowledge base, Q&A and r/careerguidance sets	download	∼17k rows	Text for the planned chatbot and for explanations in the parents' report (not training data for the classifier)
Ten Kaggle career-prediction sets	download	180–21,273 rows each	Evaluated as extra training data and **excluded** (Section **??**)

2.2 Collection details

- **APIs.** ILOSTAT data was queried through its SDMX endpoint (`sdmx.ilo.org/rest/data/ILO,<dataflow>/PAK...`) with labels included. The World Bank data comes from `api.worldbank.org/v2/country/PAK/indicator/<id>`.
- **Scraping.** Rozee.pk search pages embed the job list as JSON. The scraper walked all 550 result pages (20 jobs each), only on paths allowed by the site's `robots.txt`, with a 1.5 s delay between requests. It stored the full JSON of every posting and a flattened CSV. Scraped postings are kept out of the public repository and used only for aggregate statistics.
- **Survey.** We designed a Google Form (`survey/pakistan_student_survey.pdf`) that repeats the 48 RIASEC and 10 TIPI items word for word, so responses can be merged with the training data. It adds questions on the field the student wants, the field their parents want, who decides, and perceived family pressure. It collects no names or emails. The form is live, but no responses had arrived at the time of writing, so it is not yet part of the analysis.
- **Storage.** Raw data is too large for GitHub and is shared through Google Drive (link in the README). `.gitignore` excludes `data/`.

2.3 Supplementary datasets evaluated and excluded

To enlarge the training data we collected ten Kaggle career-prediction datasets. Each maps student attributes (skill ratings, subject marks, interests) to a recommended job. Before merging any of them, we tested whether their labels can be learned at all (`evaluate_supplementary.py`). A gradient-boosting classifier with 3-fold cross-validation predicted each dataset's own target from its own features, and a usable dataset must beat both random guessing and always predicting the most common class.

Table 2: Learnability test of the supplementary datasets (3-fold CV accuracy).

Dataset	Rows	Classes	Random	Majority	**Model**
`roo_data.csv`	20,000	34	2.9%	6.1%	**4.0%**
`dataSet.csv`	20,000	7	14.3%	28.8%	**23.6%**
`Career-prediction-data.csv`	6,901	12	8.3%	9.1%	**8.4%**
`career_path_in_all_field.csv`	9,000	90	1.1%	1.5%	**0.9%**
`career_dataset_large.xlsx`	5,000	12	8.3%	8.6%	**8.1%**
`output.csv`	1,000	10	10.0%	11.1%	**9.9%**
`Course_Job_Recommendation.csv`	21,273	45	2.2%	9.0%	**100%**
`cs_students.csv`	180	33	3.0%	10.6%	**78.3%**
`AI-based Career Rec. System.csv`	200	32	3.1%	10.5%	**25.4%**
`career_compute_dataset.csv`	7,525	–	–	–	49 labels

- **Random labels.** Six datasets score at chance level, which shows their labels were generated independently of the features (synthetic data). Training on them teaches a model noise. This explains an earlier experiment of ours that combined their categorical features and reached only 36.5% accuracy, with 0% recall for Law and Pure Sciences.
- **Target leakage.** `Course_Job_Recommendation` scores 100% because columns such as the job's role category and functional area are derived from the job title it is meant to predict. It also uses the completed degree as an input, the reverse of our task.
- **Too small or unlabelled.** `cs_students.csv` has real signal, but only 180 rows, almost all CS careers. A second copy of the same file was found and deleted. `career_compute` labels only 49 of its 7,525 rows.

All ten are kept in the shared data folder under `raw/excluded/` so the decision can be re-checked. A student-performance dataset (exam scores) was also excluded as out of scope.

3 Phase 2: Cleaning

3.1 RIASEC training data (`clean_riasec.py`)

The raw file has 145,828 rows. Figure **??** shows the missing values after recoding, and Figure **??** the two main outlier problems.

Missing values. The dataset uses `0` for "not answered" in every Likert and demographic column, even though the codebook does not define it. We recoded 0 to missing.

- Up to 4 of the 48 interest items may be missing. Each gap is filled with the person's median on the *same* scale, which preserves their profile (11,995 cells imputed).
- Respondents missing more than 4 items are dropped.
- Missing TIPI items get the column median.
- Missing categorical demographics (urban, native language, education) become an explicit "unknown" level instead of being guessed.

Invalid responses.

- **Over-claimers:** the vocabulary checklist contains three non-words. We drop anyone who claimed to know two or more of them. One is tolerated, because 20% of respondents ticked at least one, and is kept as a feature (`fake_word_flag`).
- **Speeders:** people who answered all 48 items in under 60 seconds are dropped.
- **Straight-liners:** people who gave the same answer to every item are dropped.

Outliers.

- **Age:** contains typed garbage up to 2,147,483,647. Ages outside 13–80 are dropped.

- **Family size:** contains values up to 999,999,999. Values above 20 are set to missing and median-imputed.
- **Time on test:** has a long tail (tabs left open for days). It is capped at the 99th percentile (≈66 minutes).

Duplicates. Repeated submissions with identical answers, age, gender, country and major are removed.

Target label. The free-text `major` field contains 11,816 distinct spellings (Figure **??**: "psychology", "psycology", "pyschology", "CS", ...). A keyword rule set (`house_mapping.py`) maps each answer onto one of the seven houses. Rules are ordered from specific to general, so that, for example, "computer engineering" maps to CS & IT before the generic "engineer" rule fires, and "political science" maps to Social Sciences before the generic "science" rule. Non-answers ("no", "undecided", "idk") are dropped. The rules label 92.5% of all non-blank majors.

Table **??** and Figure **??** show the effect of each step.

Table 3: Rows removed at each cleaning step.

Step	Removed	Remaining
Raw download	–	145,828
Claimed ≥2 fake words	7,322	138,506
Speeders (<60 s)	272	138,234
Straight-liners	288	137,946
Age outside 13–80	74	137,872
>4 interest items skipped	212	137,660
Duplicate submissions	2	137,658
No usable major (blank or non-answer)	56,402	**81,256**

3.2 Pakistan data (`clean_pakistan.py`)

- **ILOSTAT:** the nine indicator files are combined into one long table. Codes and labels are split apart, annual and quarterly values are separated, and 4,985 empty or duplicate observations are removed (31,969 rows remain). Average earnings by ISIC sector are then mapped to houses as a proxy, e.g. Q (health) → Medicine & Health and J (information & communication) → CS & IT.
- **World Bank:** the long table is pivoted to year × indicator and empty years are dropped.
- **Rozee.pk:**
  - 1,405 postings repeated with the same title, company and city are removed. They are mostly re-posts aggregated from other boards. 9,580 postings remain.
  - Experience text is parsed into years ("Fresh" becomes 0).
  - The minimum degree is found in only 1,833 structured fields, so it is also extracted from the description text with regular expressions, giving a level for 7,589 postings.
  - Job titles are mapped to houses with the same rule set. 7,469 postings are mapped; the rest are mostly non-degree jobs such as drivers and guards.

4 Phase 2: Feature Strategy (`build_features.py`)

4.1 Engineered features

Table 4: Model inputs (30 columns after encoding).

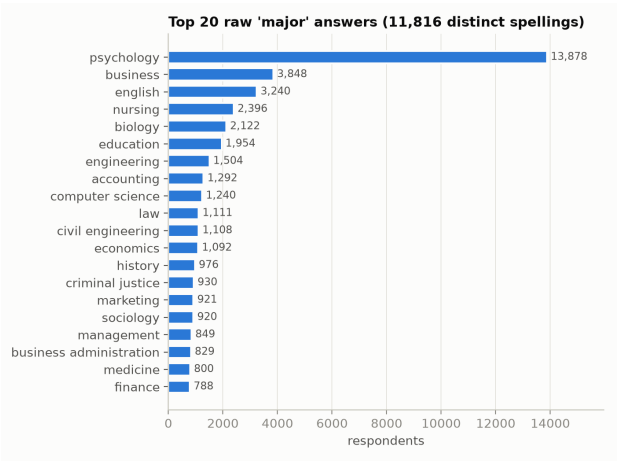
Feature	Construction	Treatment
RIASEC scores (6)	Mean of the 8 items of each scale (1–5)	StandardScaler
Big Five (5)	TIPI pairs, second item reverse-scored: $(x_a + (8 - x_b))/2$	StandardScaler
`vocab_score`	Number of the 13 real words known	StandardScaler
`differentiation`	Highest minus lowest RIASEC score (how clear-cut the profile is)	StandardScaler
`familysize`	Children of the respondent's mother	StandardScaler
`age`	Right-skewed (median 21)	$\log(1 + x)$, then StandardScaler
`education`	$1 < 2 < 3 < 4$, unknown $= 0$	Ordinal encoding, scaled
`holland_code`	Letter of the highest RIASEC score	One-hot (6)
`urban`, `engnat`	Childhood area, native English speaker	One-hot incl. "unknown"
`fake_word_flag`	Claimed exactly one non-word	Binary, passthrough
`house` (target)	7 classes	Label encoding (0–6)

4.2 Scaling, encoding and leakage

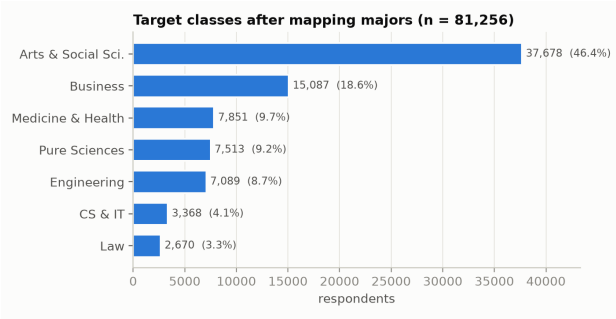
- **Pipeline.** All transformations are one scikit-learn `ColumnTransformer`. Z-score standardisation was chosen over min–max scaling because the scale scores are roughly bell-shaped (Figure **??**) and the planned linear and neural models benefit from zero-mean inputs.
- **Split and leakage.** The data is split 80/20, stratified by house (65,004 training, 16,252 test). The scalers and encoders are fitted on the training split *only*, then applied to the test split, so no test-set statistics leak into training. The fitted pipeline is saved for use in the web app.
- **Class imbalance.** Arts, Humanities & Social Sciences holds 46.4% of respondents and Law 3.3%, a 14:1 ratio (Figure **??**). We pre-computed balanced class weights (0.31 for Arts up to 4.35 for Law) for Phase 3, and will report macro-F1 rather than accuracy alone.
- **Ethics.** Gender, religion, race, sexual orientation and country are *not* model inputs. Gender and a Pakistan flag are stored beside the split only for the Phase 4 bias check.

5 Exploratory Data Analysis

5.1 Training data

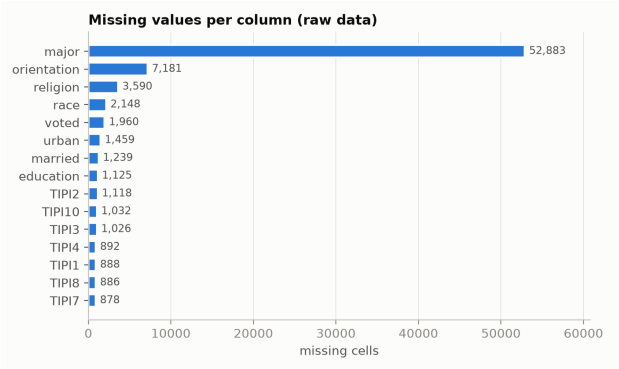


(a) Raw free-text majors before mapping.

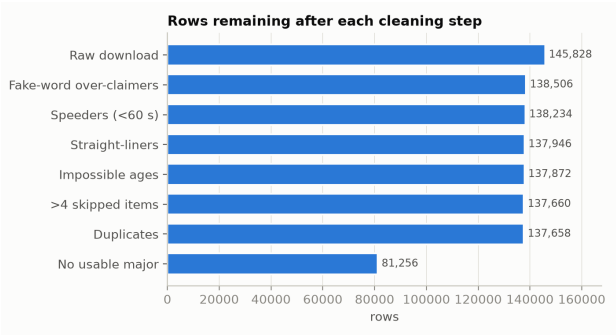


(b) Classes after mapping: 14:1 imbalance.

Figure 1: From 11,816 messy spellings to seven target classes.



(a) Missing values per column (raw).



(b) Rows remaining after each step.

Figure 2: Data quality before cleaning. Most missing majors are blank answers. Orientation, religion and race are missing more often but are not used.

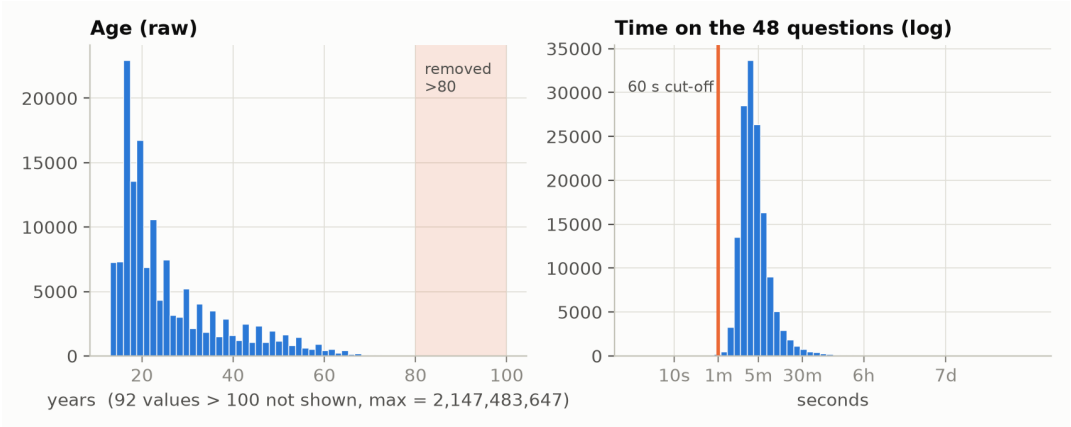


Figure 3: Outliers. Left: ages above 80 (and 92 values above 100, up to $2^{31} - 1$) are removed. Right: time spent on the test. Answers under 60 s (orange line) are treated as speeders.

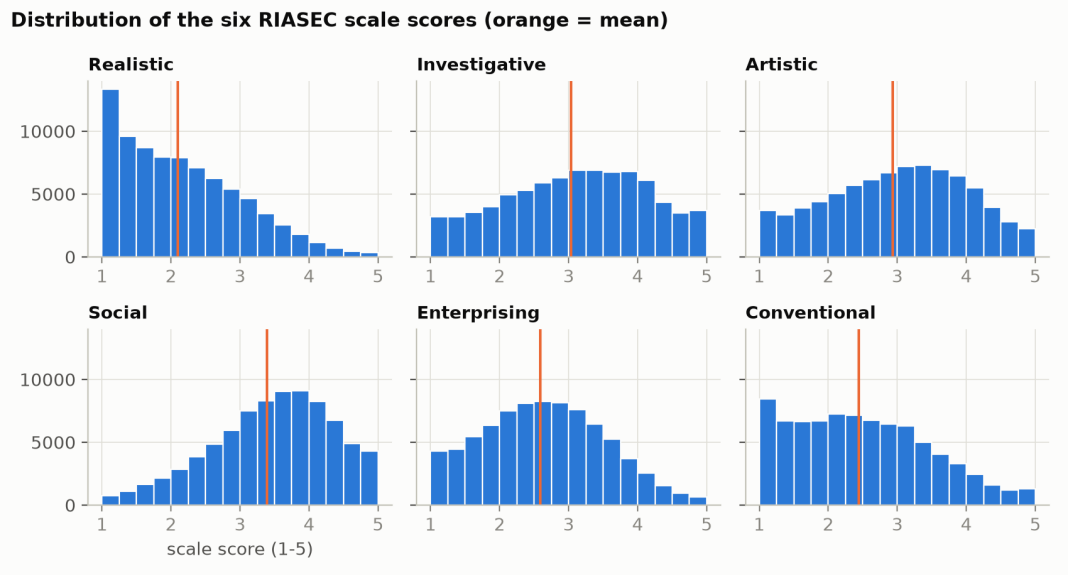


Figure 4: The six RIASEC scale scores. Social and Investigative lean high and Realistic and Conventional lean low. This reflects the sample (two thirds women, many psychology majors), not the population, which is one more reason for class weighting.

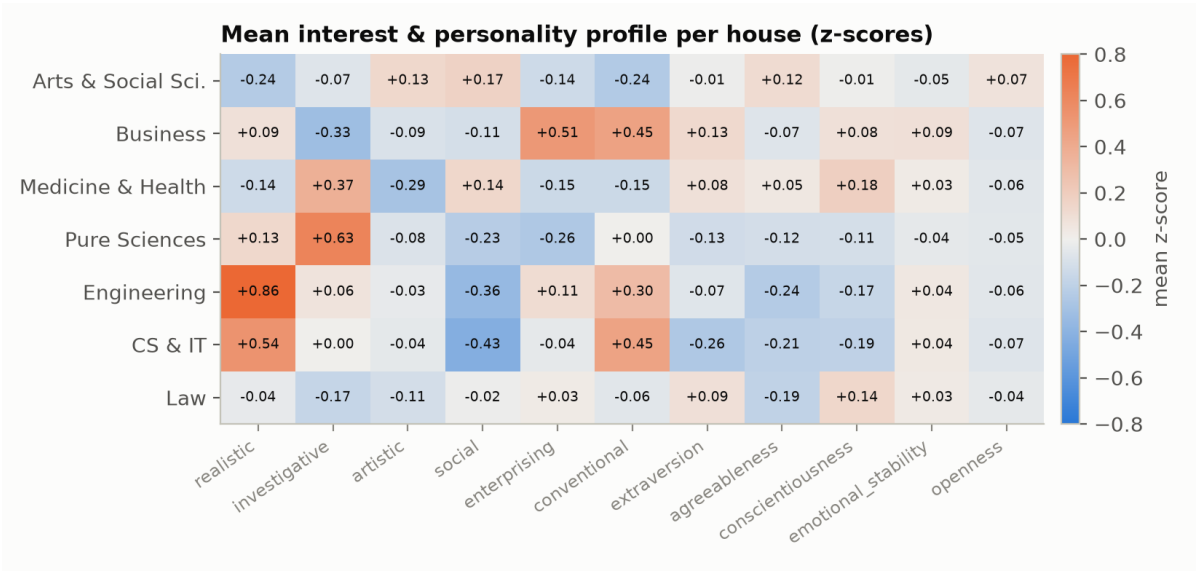


Figure 5: Mean profile of each house in z-scores. The data reproduces Holland's theory. Engineering and CS are highest on Realistic, Medicine and Pure Sciences on Investigative, Business and Law on Enterprising, and Arts & Social Sciences on Social. The overlaps between pairs of houses are why a single-rule system is not enough.

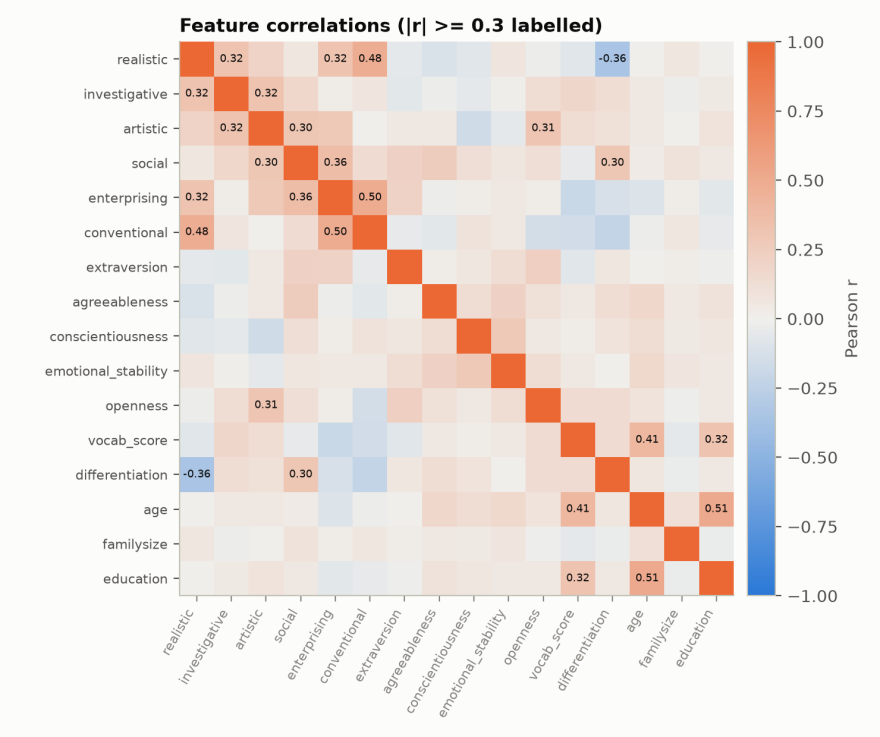
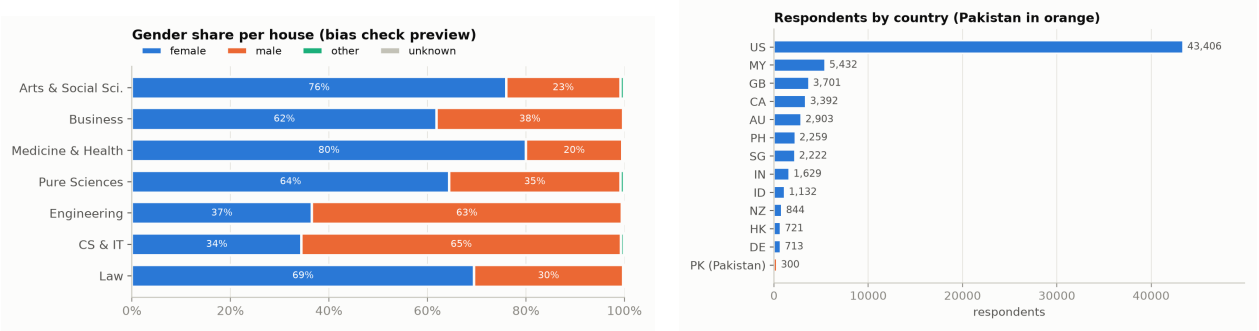


Figure 6: Feature correlations. The strongest pairs are age–education ($r = 0.51$), Enterprising–Conventional (0.50) and Realistic–Conventional (0.48), matching the adjacency of these types on Holland's hexagon. No pair is high enough ($|r| > 0.8$) to require removing a feature.



(a) Gender share per house.

(b) Country of respondents.

Figure 7: Two sources of bias to address in Phase 4. (a) Engineering and CS are about two-thirds male, so a model could learn to recommend them less to women. Gender is therefore excluded from the inputs and per-gender metrics will be reported. (b) 53% of respondents are from the US and only 300 are from Pakistan, a domain shift our own survey is designed to measure.

5.2 Pakistan statistics

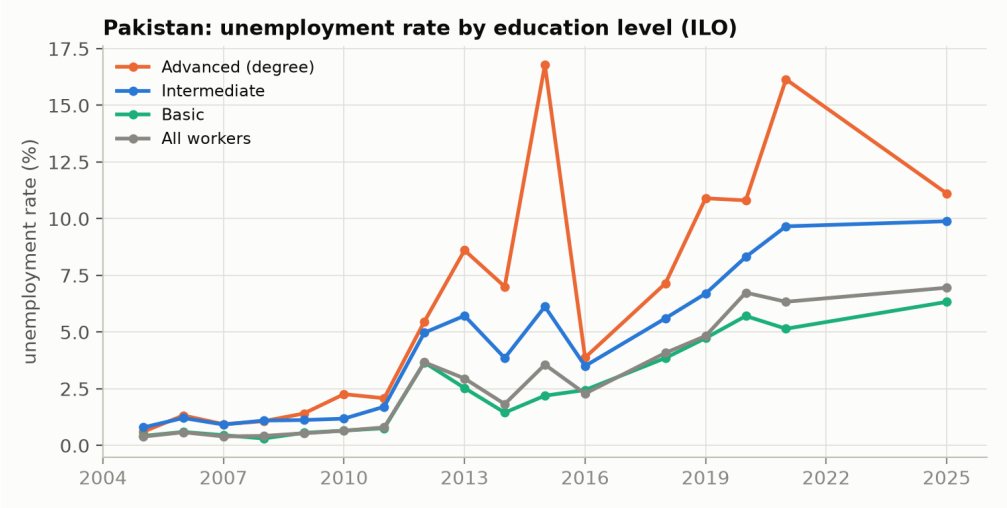
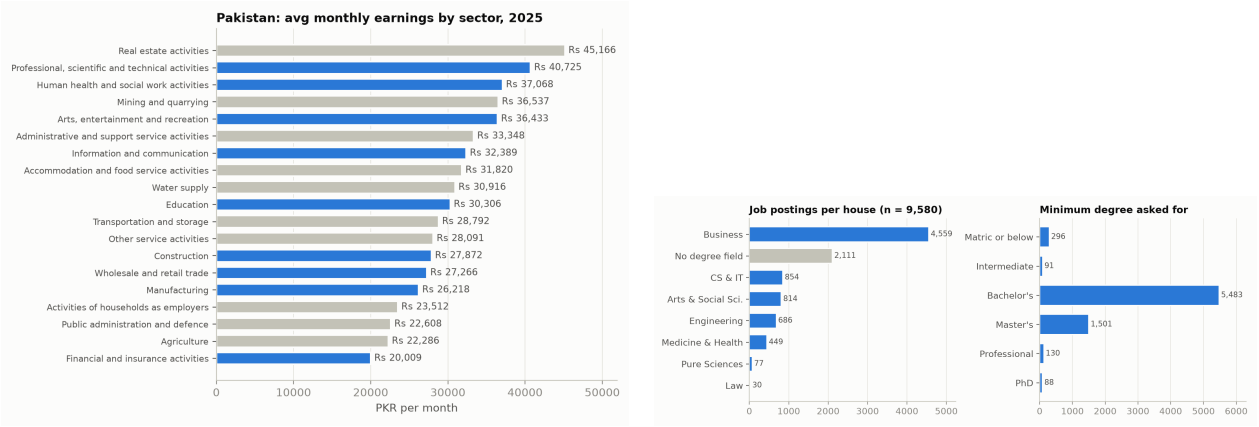


Figure 8: Unemployment rate by education level in Pakistan (ILOSTAT, Labour Force Survey). Since 2012, people with a degree have consistently had *higher* unemployment than the workforce as a whole, reaching 11.1% in 2025. The jumps in 2015 and 2016 coincide with changes in survey rounds and should be read with care.



(a) Average monthly earnings by sector, 2025.

(b) Rozee.pk postings by house and degree asked for.

Figure 9: (a) Sector averages include all employees in a sector (a hospital's cleaners as well as its doctors), so they are a coarse proxy for a degree's value. The low figure for finance in particular needs checking against occupation-level data in Phase 3. (b) Business roles dominate current postings (sales and marketing), and 72% of postings that state a degree ask for a bachelor's.

6 Preliminary Baseline (preview of Phase 3)

To check that the prepared data supports the task, we trained quick baselines on the split from Section 4 (`baseline_models.py`). Because the app will show every student a ranked list of houses, we report top-k accuracy (is the true house among the k most probable?) next to ordinary accuracy and macro-F1.

Table 5: Baselines on the held-out test set (16,252 students, 7 houses).

Model	Top-1	Top-2	Top-3	Macro-F1
Always predict the largest house	46.4%	–	–	–
Logistic regression, class-balanced	36.8%	55.1%	71.4%	0.30
Gradient boosting, 30 features	53.0%	73.0%	83.4%	0.28
Gradient boosting, 30 features + 58 raw answers	**56.6%**	**76.2%**	**86.0%**	**0.35**

- **Top-1 accuracy alone is misleading.** Always predicting Arts & Social Sciences already scores 46.4%. Class balancing lowers top-1 accuracy (36.8%) but raises macro-F1, because the model stops ignoring Law and CS.
- **Interests predict a field only moderately.** People with the same interest profile study different subjects, so a single-answer accuracy of 70–80% is not realistic from this data. Restricting training to graduates did not change the result. The best model's top-3 list contains the student's actual field **86%** of the time, which supports presenting the top three houses with probabilities rather than a single verdict.
- **Where further gains must come from:** new signals rather than more of the same data. These are the task-based career simulations planned for the app, the Pakistani survey, and tuning of the gradient-boosting and neural models in Phase 3.

7 Challenges and Limitations

- **Label noise.** Majors are self-reported free text, and some respondents are students who may still switch. The keyword mapping is transparent and testable, but imperfect.
- **Domain shift.** The training data is mostly from the US and the users are Pakistani. Only 300 Pakistani rows exist. Our survey is the planned fix, but it has no responses yet.
- **Proxy statistics.** No public dataset gives earnings per *degree* in Pakistan, so sector-level ILO data and job postings stand in for it. PBS microdata, which must be requested, would allow degree-level figures.
- **Salaries on Rozee.pk** are hidden behind a login for most postings, so salary comes only from ILO data.

8 Next Steps (Phases 3–4)

Train and compare logistic regression, random forest, gradient boosting and an MLP on the prepared split with class weights. Report accuracy, macro-F1 and a confusion matrix. Run the bias check by gender and on the Pakistani subset. Build the hosted web app and report generator.

Repository

`data_scripts/collect_data.py`	downloads all raw sources, writes the collection log
`data_scripts/house_mapping.py`	free text → house rules
`data_scripts/clean_riasec.py`	missing values, validity, outliers, duplicates, labels
`data_scripts/clean_pakistan.py`	ILO, World Bank and Rozee.pk cleaning
`data_scripts/build_features.py`	feature engineering, scaling, encoding, split
`data_scripts/eda.py`	all figures in this report
`data_scripts/evaluate_supplementary.py`	learnability test of the Kaggle datasets (Table **??**)
`data_scripts/baseline_models.py`	preliminary baselines (Table **??**)

References

[1] J. L. Holland, *Making Vocational Choices: A Theory of Vocational Personalities and Work Environments*, 3rd ed. Psychological Assessment Resources, 1997.

[2] S. D. Gosling, P. J. Rentfrow and W. B. Swann, "A very brief measure of the Big-Five personality domains," *Journal of Research in Personality*, 37, 504–528, 2003.

[3] Open Psychometrics, RIASEC raw data (collected 2015–2018), `http://openpsychometrics.org/_rawdata/`.

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[5] International Labour Organization, ILOSTAT SDMX API, `https://sdmx.ilo.org`.

[6] World Bank, World Development Indicators API, `https://api.worldbank.org`.

[7] Rozee.pk job search, `https://www.rozee.pk`, accessed 18 October 2026.